

Examination for the PhD, EdD, EngD, BusD and MSc and MLitt Degrees: notification of Degree Committee decision



This form should be completed and returned to Student Registry alongside the Examiners' independent reports and joint report.

Candidate Name	Robert William Hickman	
Candidate USN*	303969390	
Degree sought	PhD	
Degree Committee	Faculty of Biology	
College	Darwin College	
Date of Degree Committee meeting	11 May 2016	
Date of viva voce examination*	22-04-26	

Degree Committee decision

At a meeting of the Degree Committee held on the date detailed above, a quorum being present, and following consideration of the Examiners' reports, the following decision was made with regards to the student detailed above (tick one option as appropriate):

A	To provisionally award the Degree for which the student is a candidate. Final approval is conditional on submission of a hardbound and electronic copy of the thesis (not required for MLitt or MSc Degrees)	
1	Without correction	
2	Subject to only minor, or straightforward, corrections to be undertaken to the satisfaction of one or more of the Examiners.	x
3	Subject to more substantial, or less straightforward, corrections to be undertaken to the satisfaction of one or more of the Examiners.	

For A2 or A3 above, state which Examiner(s) will be responsible for confirming corrections:

Internal

B	Not to award the Degree for which the student is a candidate but to:	
1	Permit the student to revise and resubmit the thesis and take a fresh examination for the degree sought	
2	Permit the student to revise and resubmit the thesis and take a fresh examination for the degree sought <u>and</u> offer the opportunity to instead accept a MLitt or MSc Degree without resubmitting the thesis for examination, but subject to correction if required to be undertaken to the satisfaction of one or more of the Examiners (doctoral candidates only).	

*Please ensure this is completed accurately, as failure to do so may delay the processing of this form.

For B2 above, indicate whether corrections are required and if so, state which Examiner(s) will be responsible for confirming corrections.

Corrections required? Major / Minor / None

Examiner responsible for confirming corrections: Internal / External / Both

C	Not to award the degree sought but to recommend to the Postgraduate Committee that:
1	The degree of MLitt or MSc Degree be awarded without resubmitting the thesis for examination, but subject to correction if required to be undertaken to the satisfaction of one or more of the Examiners (doctoral candidates only).
2	No degree be awarded, but the candidate be offered the opportunity to revise and resubmit for an MLitt or MSc Degree (doctoral candidates only)
3	No degree be awarded, and the candidate not be offered the opportunity to revise and resubmit work for any Degree (outright fail)

For C1 above, indicate whether corrections are required and if so, state which Examiner(s) will be responsible for confirming corrections.

Corrections required? Major / Minor / None

Examiner responsible for confirming corrections: Internal / External / Both

Degree Committee Secretary/Chair: Julie Davies



Date: 12 May 2026

Examiner details

Internal Examiner	External Examiner
Professor Vincent Mak	Professor Elliott Ludvig
Judge Business School Trumpington Street Cambridge CB2 1AG UK v.mak@jbs.cam.ac.uk	Department of Psychology Coventry CV4 7AL UK E.Ludvig@warwick.ac.uk

FORM PhD2
for use on first examination of a
Ph.D. thesis



Examiners' joint report and recommendation

(to be completed *after* the oral examination)

Candidate (<i>Block Capitals</i>)	ROBERT WILLIAM HICKMAN	
Examiners (<i>Block Capitals</i>)	VINCENT MAK	Internal
	ELLIOT LUDVIG	External

Date of oral examination	22 April 2026
Format of oral examination (<i>select one</i>)	☒ In-person
	☐ Video conference – candidate and Examiners all in separate locations
	☐ Video conference – Examiners co-located, candidate in separate location
	☐ Video conference – candidate co-located with one Examiner, other Examiner in separate location

Research Impact Statement (<i>select one</i>)	☒ Research Impact Statement was not submitted by the candidate
	☐ Research Impact Statement was taken into account in the assessment

Thesis title (<i>select one</i>)	☐ Thesis title is approved
	☐ Thesis title requires amendment as detailed in our written joint report

Report - Please type your **Joint Report** of the oral examination below. Both Examiners should sign and date this form; if you prefer to attach the report on a separate sheet(s), **please also sign and date** the sheet.

The student defended the work admirably and could handle the examiners' questions well. The student is advised to perform more data analysis and discuss further the relationship of the work to previous related research. During the defence, the student demonstrated competence to follow up on these revisions.

Typed signatures will not be accepted.

	Internal Examiner	External Examiner
Signature		
Date	22 April 2026	22 April 2026

This form **will be made available to the candidate and their supervisor** when the result of the examination is communicated to the candidate.

Please turn over and complete the **Recommendation** form.

Joint Recommendation

Please indicate your joint recommendation by circling the appropriate number on the table below. You should refer to the *Guide for Examiners* for guidance.

The thesis is satisfactory for the award of the Ph.D. Degree or is expected to be satisfactory after correction										
1	Approve for the Ph.D. Degree without correction									
2	Conditionally approve for the Ph.D. Degree subject to minor corrections to be completed within 3 months Please indicate which Examiner(s) will be responsible for checking the corrections: <table border="1"> <tr> <td>Internal ☒</td> <td>External ☐</td> <td>Both Examiners ☐</td> </tr> </table> If corrections were satisfactorily completed immediately after the viva please complete and include the form PhD3.	Internal ☒	External ☐	Both Examiners ☐						
Internal ☒	External ☐	Both Examiners ☐								
3	Conditionally approve for the Ph.D. Degree subject to major corrections to be completed within 6 months Please indicate which Examiner(s) will be responsible for checking the corrections: <table border="1"> <tr> <td>Internal ☐</td> <td>External ☐</td> <td>Both Examiners ☐</td> </tr> </table>	Internal ☐	External ☐	Both Examiners ☐						
Internal ☐	External ☐	Both Examiners ☐								
The thesis requires revision such that we are unable to recommend award of the Ph.D. Degree without further examination of a revised thesis										
4	Allow to revise and resubmit the thesis for examination for the Ph.D. Degree within 12 months									
5	Allow to revise and resubmit the thesis for examination for the Ph.D. Degree within 12 months with the option for the candidate to instead accept conditional approval of the M.Litt. or M.Sc. Degree without resubmitting the thesis for further examination, subject to: <table border="1"> <tr> <td>a) no corrections required (unconditional approval)</td> <td>☐</td> </tr> <tr> <td>b) minor corrections to be completed within 3 months</td> <td>☐</td> </tr> <tr> <td>c) major corrections to be completed within 6 months</td> <td>☐</td> </tr> </table> Please indicate which Examiner(s) will be responsible for checking any corrections: <table border="1"> <tr> <td>Internal ☐</td> <td>External ☐</td> <td>Both Examiners ☐</td> </tr> </table>	a) no corrections required (unconditional approval)	☐	b) minor corrections to be completed within 3 months	☐	c) major corrections to be completed within 6 months	☐	Internal ☐	External ☐	Both Examiners ☐
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b) minor corrections to be completed within 3 months	☐									
c) major corrections to be completed within 6 months	☐									
Internal ☐	External ☐	Both Examiners ☐								
The candidate is not to be approved for the Ph.D. Degree and not allowed to submit a revised thesis for the Ph.D. Degree (recommendations should be used only in exceptional cases on a first examination)										
6	Not allow revision of the thesis for the Ph.D. Degree but conditionally approve for the M.Litt. or M.Sc. Degree without resubmitting the thesis for further examination, subject to: <table border="1"> <tr> <td>a) no corrections required (unconditional approval)</td> <td>☐</td> </tr> <tr> <td>b) minor corrections to be completed within 3 months</td> <td>☐</td> </tr> <tr> <td>c) major corrections to be completed within 6 months</td> <td>☐</td> </tr> </table> Please indicate which Examiner(s) will be responsible for checking any corrections: <table border="1"> <tr> <td>Internal ☐</td> <td>External ☐</td> <td>Both Examiners ☐</td> </tr> </table>	a) no corrections required (unconditional approval)	☐	b) minor corrections to be completed within 3 months	☐	c) major corrections to be completed within 6 months	☐	Internal ☐	External ☐	Both Examiners ☐
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1. The thesis makes a compelling empirical point about the relationship between dopamine neurons and utility, with many alternative possibilities considered and ruled out. As discussed during the viva, the paper could use more connection to a broader theoretical angle (especially in the first and last chapters). There is, of course, mention of reward prediction errors and reinforcement learning, but there is less engagement than I would expect with those theories, which of course are very prominent in interpreting the role of dopamine neurons in learning and choice. How do the current results interact with that theory—a number of results seem very challenging, including the very strong responses to clearly predicted outcomes? What about alternative theories in that space, such as the newer work from the Namboodiri lab on causal learning or other newer developments? Some of the more interesting bits that speak to theory are relegated to the appendices.

2. Throughout, the paper uses frequentist statistics, but with a unit of analysis seemingly at the level of trials or neurons, not individuals, which makes sense given an $n = 2$ monkeys. This raises issue related to the independence of the data points, which may inflate significance. As discussed in the viva, a mixed-model approach would be suitable in these instances where there is internal hierarchical structure to the sampled data (monkey, neuron, trial, etc.). The corrections should add such a mixed-effects model for the most central claim in the thesis—that dopamine response correlates with bid.

3. In addition, one of the key claims in Chapter 4 that there is no difference in neuronal responding for different magnitude outcomes when the bids are the same (dissociating utility from magnitude). This analysis should have the evidence for the null quantified with a suitable Bayesian test.

4. One of the nice variations of the core task in the thesis was the inclusion of single-fractal sessions with the medium-value fractal. The interpretation here does not fit that strongly with the apparent results. The thesis downplays the difference between valuation/utility between single and multiple-fractal sessions, but one monkey shows a very clear effect here ($d = 0.46$ is not small in typical social science terms). In the broader behaviour economics literature, relative valuation is very commonly observed, and there are a number of theories that are predicated exactly on this type of relative valuation. I think this aspect of the data should be taken more seriously. In addition, the related dopamine response should be examined, as this could be extremely strong evidence for the core contention that dopamine corresponds to utility rather than reward.

Other points (chapter-by-chapter):

Ch 1.

- Bottom p. 30 way overstates the utility of EUT in modern studies of decision-making. Beh Econ has mostly veered away from them due to the sheer number of violations. The base model

these days is CPT (which is also wrong in lots of ways), but people rarely start from EUT anymore, at least not in JDM.

- P. 32. Rescorla-Wagner does not really have a cause-effect structure. More associative. Not even predictive (strictly). RW is only trial-level.

Ch 3: Training monkeys in a BDM

- Fig 3.4. What determined the end of a session? Should be mentioned.

- Effect of starting bid and total liquid are opposite across the two monkeys? Not enough discussion of this point and what it means for interpreting the results.

- p. 111. Unclear what the tie fighters are in this plot? Where is the BDM vs choice comparison? Missing blue tie fighter. Need to expand the range of the plot to accommodate? Caption also hard to parse which is used as utility (BDM or Choice) and which is used to compare.

Chapter 4

- Figure 4.5. I was confused as to where the other putative dopamine neurons went? They were included in the upper plot and not the lower plot? Were they not dopamine neurons? Did they just not show responses to unpredicted rewards? Are these the ones that were only tested at least 5 times? But why were only some tested 5 times? This needs some additional clarification (and was discussed in viva).

- Fig 4.11 is the crux of the thesis. Dopamine encodes upcoming bid. But, this is confounded by value of the upcoming reward? Figure 4.13B seems like it might be better fit by a three-line spline. As is done in Fig 4.17, would be more convincing to look at each reward level separately and show that the dopamine predicts bids even for identical predicted magnitudes. Should foreshadow this.

- p. 139. Why bin the bids? Throwing away data. Why not just use the raw bids as a continuous predictor? Needs justification or inclusion in the primary analysis (see point 2 at the top)

- Fig 4.16. p. 144. What is the inverted U shape? That is a very notable feature of the plot that needs remarking on and some speculation as to why it's there.

- Table 4.3 seems to have an error and the same data are reported on 3 different rows

- Fig 4.27A seems to show that medium is above high?? And all seem to go up, yet the text on p. 163 indicates that only medium and high go up? Stats and description don't seem to align with figure on this one. Same with Fig 4.27B. This discrepancy needs confirmation of what the real data showed, and, if correct, then some additional explanation is in order. Same again in Fig 4.28, where Monkey U has medium above high? Not significant but also not aligned with fig.

App A

- p. 209. The logit choice rule is just softmax, which has long been used in other areas? Might be worth a mention

Smaller things:

- Eq 1.5 has error (missing first preference term).

- Bot p. 28. Should be Eq 1.2, not 1.3
- p. 33. Missing bracket after robinson and Berridge (first par)
- Fig 1.2 is confusing in that it mixes real data (a) with simulated (b and c) without clear visual demarcations.
- p. 46. Misplaced bracket
- p. 48. Line 5. Less than sign misplaced (in subscript)
- p. 50. Intro does not really discuss PT and loss aversion (move info from appendix?)
- p. 59. Odd to insert as (very long) bullet points, rather than weave as proper text.
- p. 68. Missing bracket at end of Singh ref
- p. 70. Again odd to use such very long bullet points instead of plain narrative text
- p. 73. Table 3.1 caption. “Exuded” instead of excluded
- p. 75. Dashed lines “indicate”
- p. 76. Table 3.2 (end of penultimate paragraph—missing Table)
- p. 80 “finial” instead of “final” about 6 lines from bottom
- p. 87. Fig 3.10. caption. “Spit” instead of “split”
- Axis labels on Fig 3.10 are pretty unreadable. Stream of numbers. Needs cleaner organization for so much data.
- Fig 3.12 top row. Y-axis labels are missing or on wrong side.
- p. 92. Footnote 5. Numbers are overprecise.
- p. 93. “Show and effect of” (should be an)
- p. 95. $d = 0.47$ is a medium effect, not small and considered important in psych
- p. 99. the estimates of other parameters (missing of)
- p. 108. Table 3.9. Trail instead of trial x 4
- p. 112. Is table 3.10 really supposed to be referencing figure 1.1, rather than a recent figure from the bundle experiment?
- p. 123. Wilcox.test presumably, not .text
- p. 123. Missing space between one-sided and unless
- p. 127. Fig 4.4. caption. Left and right (or blue and orange) are described backwards
- p. 127. Fig 4.4. There are no y axis scales (which are stated to differ)
- p. 145. Stray period as its own paragraph
- p. 145. 4th line from bottom. Table 4.3. (table missing)
- p. 157. Caption Fig 4.23. Biding instead of bidding.
- p. 163. Caption Fig 4.27. Did not “big” higher (instead of bid)
- p. 164. Clause starting “no tercile split” is very unclear.
- p. 181. Double semi-colon.
- p. 208. Such “binary choice” tasks (no a)
- p. 210. Fig A3 Caption. “and error term” (should be an)

FORM PhD1

*for use on first examination of a
Ph.D. thesis*



**UNIVERSITY OF
CAMBRIDGE**

Examiner's independent report and recommendation

(to be completed *before the* oral examination)

Candidate	Examiner	Date of report
Robert William Hickman	Vincent Mak	22 April 2026

1. Please attach this form to your **Independent Report** on the thesis; please sign and date the report. Please note that **your report will be made available to the candidate and their supervisor** when the result is communicated to the candidate
2. Please indicate your pre-oral recommendation by circling the appropriate number on the table below; you may wish to refer to the *Guide for Examiners* for guidance.

The thesis is satisfactory for the award of the Ph.D. Degree or is expected to be satisfactory after correction	
1	Approve for the Ph.D. Degree without correction
②	Conditionally approve for the Ph.D. Degree subject to minor corrections (3 months)
3	Conditionally approve for the Ph.D. Degree subject to major corrections (6 months)
The thesis requires revision such that I am unable to recommend award of the Ph.D. Degree without further examination of a revised thesis	
4	Allow to revise and resubmit the thesis for examination for the Ph.D. Degree
5	Allow to revise and resubmit the thesis for examination for the Ph.D. Degree or accept the M.Sc./M.Litt. Degree without further revision or examination of the thesis
The candidate is not to be approved for the Ph.D. Degree and not allowed to submit a revised thesis for the Ph.D. Degree	
6	Conditionally approved for the M.Sc./M.Litt. Degree without further revision or examination of the thesis
7	Allow to revise and resubmit the thesis for examination for the M.Sc./M.Litt. Degree only
8	Not approved nor allowed to revise the thesis for any degree - outright failure
9	I wish to reserve judgement on the outcome until after the oral examination

If the student submitted a Research Impact Statement, please confirm that you have taken this into account in your assessment:

Yes	
n/a	✓

Please **sign and date** this form and submit it, together with the other papers, to the Degree Committee office after the oral examination.

Typed signatures will not be accepted.

Signed: 	Date: 22 April 2026
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PhD THESIS REPORT (pre-viva)
(Total length to be no less than 450 words)

*The Degree Committee relies on high quality reports from examiners in order to make decisions on the award of degrees. Use of this template is **optional**, but **all reports must meet the 450 minimum word count and cover the areas detailed below.***
The report should include a technical description of the dissertation, comments on its contribution to learning, and a recommendation.

Candidate name: Robert William Hickman

Title of thesis: A Measurement of Neuronal Encoding of Subjective Value on a Moment by Moment Basis: The encoding of value by midbrain dopamine neurons in an auction-like task

Examiner's name: Vincent Mak

Examiner's address: Cambridge Judge Business School, University of Cambridge, Trumpington Street, Cambridge CB2 1AG, UK

1. Content of Thesis, Originality and Contribution to Learning (min. 200 words)

Please briefly summarise the scientific content of the thesis and consider: does the thesis take account of previously published work? Does it represent a significant contribution to knowledge through the discovery of new knowledge; the connection of previously unrelated facts; the development of new theory; or the revision of older views?

The thesis research involves a multi-component project in which the Becker-DeGroot-Marschak (BDM) auction mechanism, a well-established approach to eliciting valuations from human participants with an incentive compatible approach, was applied to eliciting valuations from two monkeys (rhesus macaques). The researchers used an ingenious set-up to make the BDM mechanism a viable tool when the experimental participants were monkeys.

Specifically, the research involved: (a) a behavioural component in which the dependent variable was primarily the measured BDM bids from the monkeys; (b) a benchmarking behavioural component in which the dependent variable was a binary choice from which valuations could be estimated, in order to compare against the BDM values; and (c) an electrophysiological component in which the monkeys' dopamine response in the experiment was measured via brain implants.

A previous, different study with similar behavioural components was reported in a published paper by co-authors from the same group of researchers (Al-Mohammad and Schultz 2022, cited in the thesis). A report of the electrophysiological component

of the present research has been published, with the student as a co-author, as Hill et al. (2024, cited in the thesis). As acknowledged by the student at the beginning of the thesis, the thesis contains the student's diligent and comprehensive analysis of the data from the experiment.

The BDM method has a long history in human behavioural research, but the student and their research group's implementation of the method among monkeys is a novelty. As noted in the third paragraph on p.156, even human participants might find it difficult to grasp the logic behind BDM, while here is a demonstration that monkeys, with sufficient training, could perform a task with a BDM design in a way that seems to be able to elicit valuations.

The three components hang together very well in exposition, proceeding from the behavioural components to the electrophysiological component. Overall, the research explored new methodological frontiers in understanding how the brain encodes values.

2. Quality of Work (min. 150 words)

Please consider whether the work is, in part, of a standard to merit publication? Is there evidence of a detailed understanding of research techniques? Does it demonstrate advanced academic enquiry? Does it reflect 3 years of full-time (5 years of part-time) study?

I appreciate the ingenious design that enabled monkeys to submit bids in a second-price auction with data consistent with incentivised behaviour. I also appreciate the amount of painstaking work the student and their collaborators put into the years-long enterprise that reflected 3 years of full-time study.

In my view, the research is of a standard that merits publication, with a detailed, hands-on understanding of research techniques demonstrating advanced academic enquiry. My view corroborates with the fact that a version of the research has indeed appeared in publication in a well-regarded academic journal (Hill et al. 2024).

In the context of a PhD thesis, I think the student can consider further elaborations of the contributions of the work. The BDM-related investigations from the student's research group are collectively highly original. The present research also makes a contribution over Al-Mohammad and Schultz (2022) from the same group with the electrophysiological component, with results showing how the well-trained monkeys' dopamine reward signals had moment-by-moment correspondence with their bids. That said, these results are in line with well-established results about dopamine reward signals, and I wonder if more can be said about any other novel aspects of the data. Currently, the related discussion seems to be largely concentrated on one page in Section 5.4. The rest of Section 5 offers some good insights that might be worth further exploration in the sense discussed here.

While both Al-Mohammad and Schultz (2022) and the present research have similar behavioural components and had the same monkey participants, there also seem to be some differences between the results. For example, as summarised on p.86, the binary choice utility results from the present research seem to be lower than those obtained from the BDM method, while they were about the same in Al-Mohammad and Schultz (2022). Beyond the brief discussion on p.86, could some meaningful potential factors be explored, if only speculatively? These might also increase the potential contributions of the behavioural component beyond a replication.

More generally, it seems that some of the results are more intriguing than straightforwardly predictable from previous research. In those cases, the student tends to offer a quick explanation (e.g., p.145 and p.155, last paragraph). Maybe some of those observations could promise new avenues of research and are worth a more detailed discussion.

3. General Style and Presentation (min. 100 words)

Please consider the structure, flow, clarity, use of language and presentation of data.

The thesis is well-written and well organised overall. I appreciate not only the two chapters that report the actual research (Chapters 3 and 4), but also the more general, background chapters (Chapters 1 and 2), the discussion in Chapter 5, as well as the Appendices. The additional content involves some reflections on utilities and values, an introduction to related neuroeconomic findings, an interesting discussion on auctions, and even along the way, a quick introduction to game theory including incomplete information games (with an illustrative example on p.189). Such discussions would have been peripheral or unnecessary for a succinct research paper, but they are appropriately woven into the structure here, demonstrating the student's in-depth understanding of the research and its different facets.

I have some specific minor comments, mostly regarding exposition:

1. It might be better to explicitly (i.e., formally) present the payoff and cost of misbehaviour functions for uniform, normal and matched normal distributions of computer bids, perhaps in an appendix. Even for uniform distribution, which should be the simplest to tackle, the presentation in Appendix E could be more complete.
2. Some of the notations could be more consistent and simplified. For example, the term "Budget" in the early part of Appendix E suddenly becomes " $B_{\max}$ ", and although it's explained during the transition that the budget is the maximum bid, why not use " $B_{\max}$ " all through and explain at the beginning that $B_{\max}$ is the budget? The multiple lines of equations in (E.7) also seem to repeat the term on the left-hand side too often.
3. Hill et al. (2024) indicate that the data and analysis codes are publicly available and specify how they can be accessed; it might be good to do the same in the thesis.

4. The ethics discussion in Hill et al. (2024) should also be incorporated in the thesis, in a form that the student deems appropriate.
5. On p.44, it is mentioned that the monkeys went on fluid restrictions on a weekly schedule for the experimentation. I wonder if there is any within-week cycle of behavioural changes that is discernible from the data?
6. There are occasional typos. Examples include p.62, beginning of Section 3.2.3: “monkeys bids”; p.100: “Mirenowicz and Schultz 1994” (need brackets around “1994”); p.157: “the starting big they wish to make”. A thorough check might be useful.

4. Summary and Recommendation (min. 50 words)

*Please summarise your overall impression and recommend an outcome. Include here one of the nine statements listed on form PhD1, **in addition to completing form PhD1 itself.***

The thesis reflects diligent, in-depth research that could advance relevant scientific understanding. The writing displays a thorough understanding of background academic knowledge as well as a state-of-the-art and hands-on engagement with the research. While I have made a number of comments as stated above in the spirit of continuous improvement, the present work is sufficiently demonstrative to merit the degree. Thus, my recommendation is “Conditionally approve for the Ph.D. Degree subject to minor corrections (**3 months**)”, as I have also stated on form PhD1.

Signed:



Date: 22 April 2026

Examiner's independent report and recommendation
(to be completed before the oral examination)

Candidate	Examiner	Date of report
Robert Hickman	Elliot Ludvig	April 21, 2026

1. Please attach this form to your Independent Report on the thesis; please sign and date the report. Please note that your report will be made available to the candidate and their supervisor when the result is communicated to the candidate
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If the student submitted a Research Impact Statement, please confirm that you have taken this into account in your assessment:		<table border="1"> <tr> <td>Yes</td> <td></td> </tr> <tr> <td>n/a</td> <td></td> </tr> </table>	Yes		n/a	
Yes						
n/a						

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Typed signatures will not be accepted.

Signed: 	Date: April 21, 2026
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PhD THESIS REPORT (pre-viva)
(Total length to be no less than 450 words)

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Candidate name: Robert Hickman

Title of thesis: A Measurement of Neuronal Encoding of Subjective Value on a Moment by Moment Basis

Examiner's name: Elliot Ludvig

Examiner's address: E.Ludvig@Warwick.ac.uk

1. Content of Thesis, Originality and Contribution to Learning (min. 200 words) Please briefly summarise the scientific content of the thesis and consider: does the thesis take account of previously published work? Does it represent a significant contribution to knowledge through the discovery of new knowledge; the connection of previously unrelated facts; the development of new theory; or the revision of older views?
See report below
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See report below
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See below

Signed: 

Date: April 21, 2026

Pre-thesis Independent Report

This thesis examined the neural basis of economic utility in monkeys. The thesis used a novel auction procedure, whereby monkeys could bid (with water) for different amounts of juice (a preferred reward) and recorded from putative dopamine neurons while monkeys performed the tasks. There were several variations of the task (e.g., with multiple reward levels and some binary choices) which together paint a compelling picture that dopamine neurons are sensitive to the immediate experienced utility of the potential rewards.

I think the strongest part of the thesis was the very careful and extensive data analysis that were detailed in Chapter 3 and 4. These very carefully broke down exactly what factors were influencing what the monkey was choosing as well as the corresponding neural responses. Nearly every time I had an objection or concern about a certain approach, the very next page would present an alternative that answered the concern. The writing was clear and fluid, though a little bit on the informal side at times. That informality would likely need to be tightened up/removed in a version for publication, but worked well in the context of the thesis. The body of work easily meets the threshold of 3-years full-time work, given the extensive training, data collection, analyses, and clearly careful thought behind many of the design choices, analyses, and interpretations.

There were a few broad areas that could use further discussion in the thesis and may necessitate corrections. I note three of them here:

1. The thesis makes a pretty compelling empirical point about the relationship between dopamine neurons and utility, with lots of alternative possibilities considered and ruled out. What I would have liked to have seen more of is a broader theoretical angle (especially in the first and last chapters). There is, of course, mention of reward prediction errors and reinforcement learning, but there is less engagement than I would expect with those theories, which of course are very prominent in interpreting the role of dopamine neurons in learning and choice. How do the current results interact with that theory—a number of results seem very challenging, including the very strong responses to clearly predicted outcomes? What about alternative theories in that space, such as the work from the Nambodiri lab on causal learning? Some of the more interesting bits that speak to theory are relegated to the appendices.

2. The general inferential statistics approach could use more justification and explanation. Throughout, the paper uses frequentist statistics, but with a unit of analysis seemingly at the level of trials or neurons, not individuals, which makes sense given an $n = 2$ monkeys. I worry a little a bit about the independence of the data points, which may inflate significance. Though common in this field (for good reasons), I also worry about the strength of inference with only a sample of two individuals—a point which is not really discussed. A mixed-model approach might have been worth considering in many instances where there is internal hierarchical structure to the sampled data (monkey, neuron, trial, etc.). In addition, one of the major claims in Chapter 4 that there is no difference in neuronal responding for different magnitude outcomes when the bids are the same (dissociating utility from magnitude) should likely have had the evidence for the null quantified with a suitable Bayesian test.

3. One of the nice variations of the core task in the thesis was the inclusion of single-fractal sessions with the medium-value fractal. The interpretation here does not fit that strongly with the apparent results. The thesis downplays the difference between valuation/utility between single and multiple-fractal sessions, but one monkey shows a very clear effect here ($d = 0.46$ is not small in typical social science terms, though see point 2 above). In the broader behaviour economics literature, relative valuation is very commonly observed, and there are a number of theories that are predicated exactly on this type of relative valuation. I think this aspect of the data should perhaps be taken more seriously.

Overall, I think that the thesis will prove satisfactory for the award of the Ph.D. Degree, and I conditionally approve for the Ph.D. Degree subject to minor corrections.